Harthik Mallichetty

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WORK EXPERIENCE

Room (*joinaroom.app*) **May 2026 – Present**
AI Engineer & Founder *Dallas, TX*

- Built a multi-model AI workspace using Next.js, TypeScript, and OpenRouter, streaming up to three LLMs (GPT, Claude, Gemini) in parallel via SSE with targeted routing, fault isolation, and server-side cancellation.
- Engineered a PostgreSQL backed project memory pipeline that extracts and persists source linked decisions, constraints and facts, enabling user confirmed context retrieval across independent AI conversations.
- Designed an LLM evaluation suite achieving 24/24 memory-recall and 26/26 cross-model continuity passes, with 0 successful prompt injections across 30 controlled adversarial tests.
- Implemented secure multi-tenant data persistence using Supabase/PostgreSQL, Row Level Security (RLS), transactional updates, and retry-safe recovery, validated through 971 automated unit and integration tests.

Prolifics **February 2023 – July 2024**
Associate Data Engineer | June 2023 – July 2024 *Hyderabad, India*

- Designed and deployed a TensorFlow-based sprint risk prediction service as a Flask REST API, giving 15+ engineering managers 1–2 weeks' advance visibility into at-risk sprints.
- Built SQL ETL pipelines integrating 250K+ Jira and QA records into datasets for risk modeling and analytics.
- Integrated risk predictions into dashboards for delivery teams of 200+ developers, with documentation and onboarding.

Data Engineer Intern | February 2023 – May 2023

- Built SQL ETL pipelines consolidating 100K+ Jira and PostgreSQL records into reusable engineering analytics datasets.
- Standardized sprint health, defect trend, and delivery KPIs and resolved data quality issues across QA and project systems, giving QA and delivery teams one shared reporting framework.

HeiDigi **March 2022 – August 2022**
Data Analyst Intern *Hyderabad, India*

- Automated KPI reporting for 20+ marketing campaigns using Python and SQL, standardizing measurement.
- Analyzed campaign experiments to find top audiences and creatives, contributing to 20% higher ROI and 18% higher CTR.

PROJECTS

Code Sentinel – AI Code Review System (*github.com/harthikrm/code-sentinel*) **March 2026**

- Fine-tuned Mistral-7B-Instruct on 143K code-review pairs using QLoRA on an NVIDIA H100, achieving 6.1% higher BERTScore F1 than GPT-4o-mini with under 1% trainable parameters.
- Deployed a FastAPI inference service on Vertex AI and integrated a GitHub Action that posts AI-generated review comments on pull requests.

Signal – Financial Intelligence (*signal.harthik.dev*) **December 2025**

- Built a financial research platform processing 800+ requests in 30 days, enabling natural-language analysis of SEC filings across 70 S&P 500 companies through a 10-step RAG pipeline and LangGraph agent.
- Engineered the data foundation on GCP: Airflow-orchestrated ingestion of SEC EDGAR filings and market data into PostgreSQL, with 3-layer dbt transformations computing 65 financial metrics per company.

EDUCATION

University of Texas at Dallas, Richardson, TX **May 2026**
Master of Science in Business Analytics and Artificial Intelligence - Scholar With Distinction GPA: 3.8/4.0

GITAM University, Hyderabad, India **April 2024**
Bachelor of Technology in Computer Science and Business Systems GPA: 8.52/10

SKILLS & ADDITIONAL

- **Languages & Databases:** Python, SQL, R, JavaScript, TypeScript, PostgreSQL, MySQL, pgvector, DuckDB
- **Data Engineering & Analytics:** ETL/ELT, Data Modeling, Airflow, dbt, Pandas, Power BI, Tableau, A/B Testing
- **Cloud & Backend:** GCP (Cloud Run, Vertex AI, Cloud SQL), Next.js, Supabase, Docker, FastAPI, Git, GitHub Actions
- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, Hugging Face, QLoRA, LangChain, RAG, LLM evaluation
- **Competitions:** ARC Prize 2026 (ARC-AGI-2) | Kaggle Gemma 4 Developer Agent Competition
- **Programs:** McKinsey.org Forward Program (2026) | IBM Data Analytics Academic Internship (2023)
- **Interests:** Golf, Poker, Chess, Fishing, Camping, Traveling, Cooking, Cricket, Soccer, Formula 1